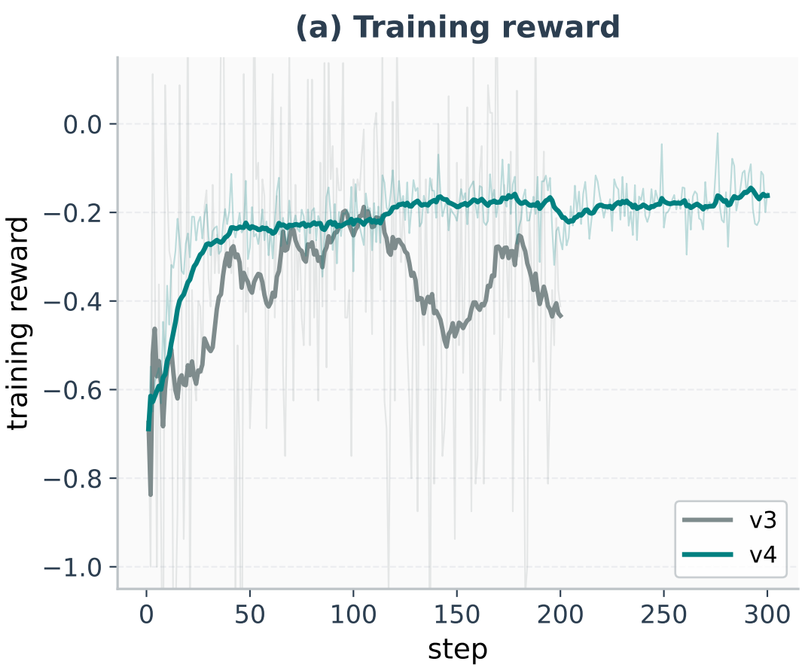
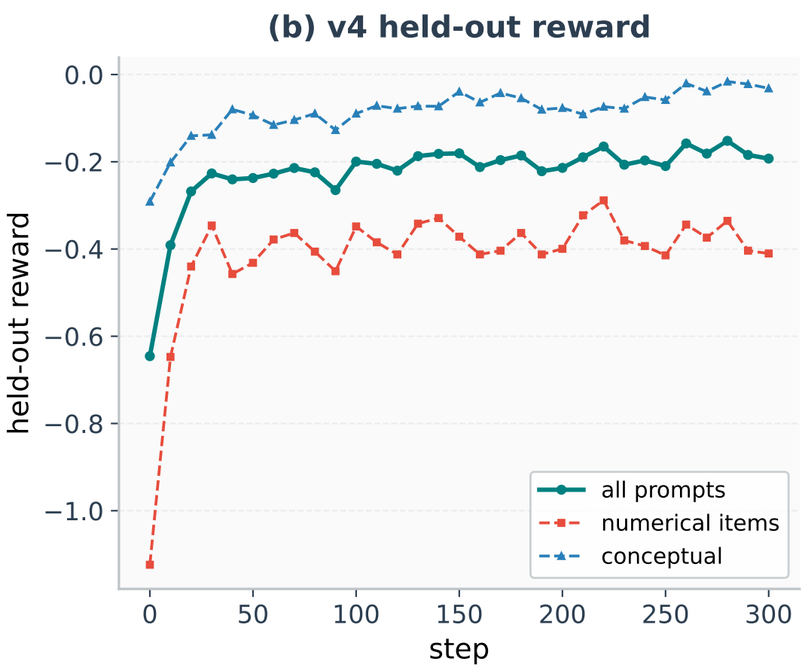
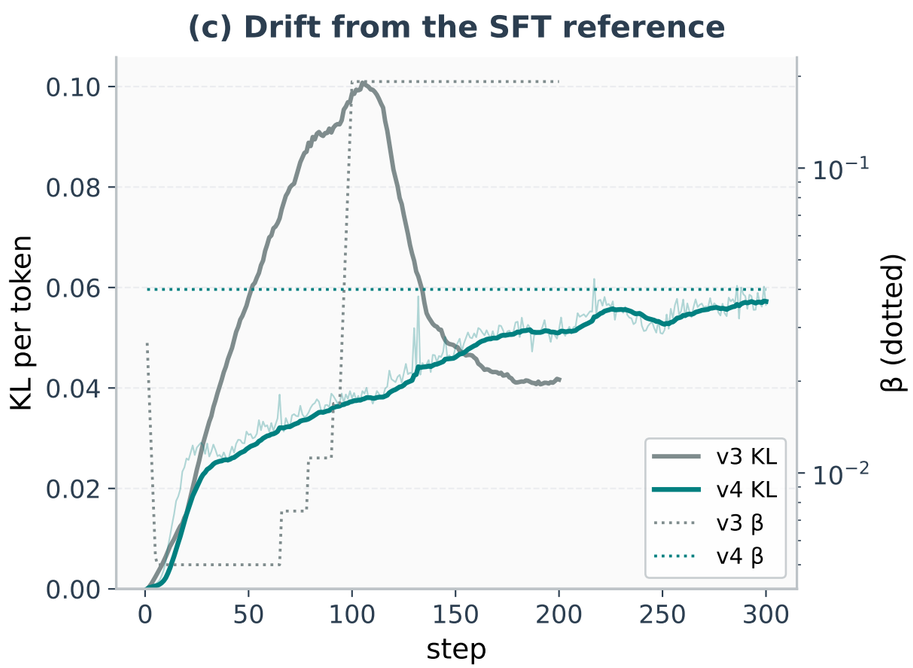
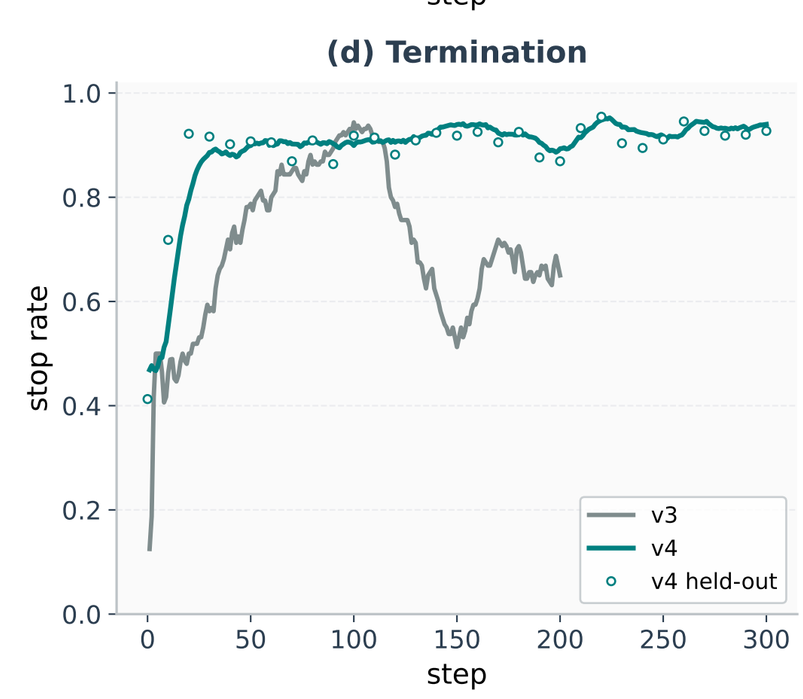
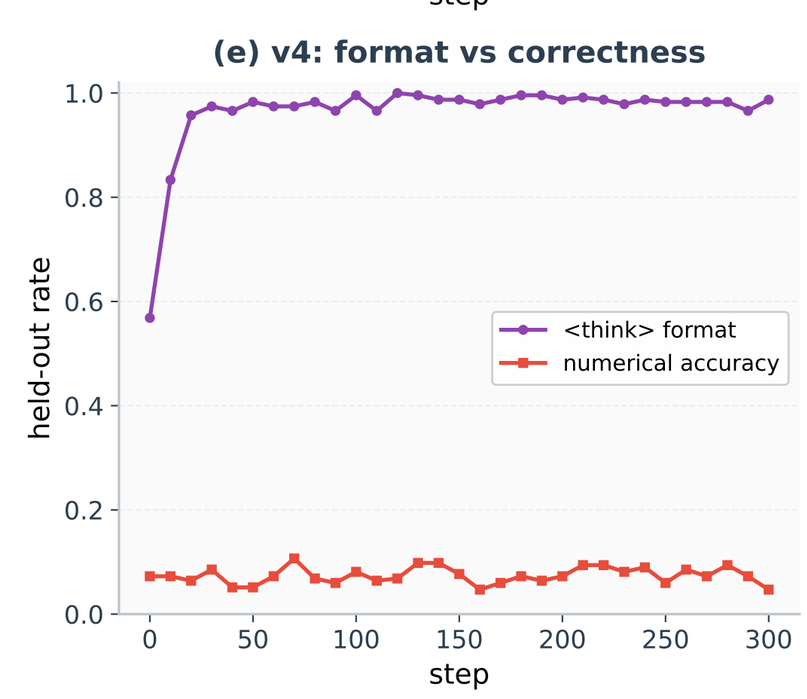
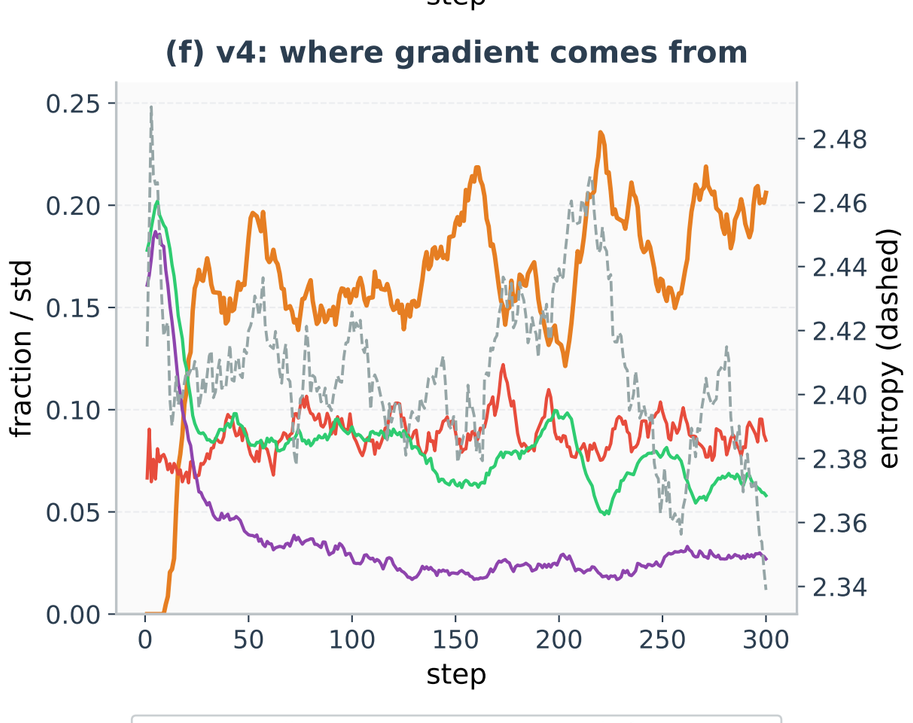

(a) Training reward
training reward
step
v3
v4


(b) v4 held-out reward
held-out reward
step
all prompts
numerical items
conceptual


(c) Drift from the SFT reference
KL per token
β (dotted)
step
v3 KL
v4 KL
v3 β
v4 β


(d) Termination
stop rate
step
v3
v4
v4 held-out


(e) v4: format vs correctness
held-out rate
step
<think> format
numerical accuracy


(f) v4: where gradient comes from
fraction / std
entropy (dashed)
step
no-gradient groups
std of numerical reward
std of format reward
std of stop reward
entropy (nats)